1 Introduction

- Most mathematical models of real-life processes involve differential equations (e.g. mechanical systems, electrical circuits, chemical reactions, biological populations, financial markets).
- An ordinary differential equation (ODE) is an equation relating values of a function ($y(t)$, say) to its derivatives, for example,

$$\frac{dy}{dt} + 6y = 0 \quad \text{or} \quad \frac{d^2y}{dt^2} - \frac{4t+1}{t}\frac{dy}{dt} + \frac{2(2t+1)}{t}y = 4e^{2t}\,.$$

- The function $y(t)$ is the **unknown function** that is to be found by solving the differential equation. Recall that a function $y(t)$ has an **independent variable** t and a **dependent variable** y.
- A differentiable[1] function $y(t)$ is a **solution of a differential equation** in an interval $I : a < t < b$, say, if the substitution of $y(t)$ into the differential equation produces an equality which is true for all $t \in I$. The interval I (which must be continuous) may be finite or infinite, i.e. $a \to -\infty$ and $b \to \infty$ are allowed.

[1] A differentiable function $y(t)$ is a function whose derivative

$$\frac{dy}{dt} = \lim_{h\to 0}\frac{y(t+h)-y(t)}{h}$$

is defined at all points in the domain. Note that $y(t)$ must be defined in order that dy/dt is defined. See the MTH1002, part 1, lecture notes.

Example 1.1 Consider a bank account in which the money grows in such a way that the increase per unit time t is proportional to the current balance $y(t)$, with a constant coefficient k, i.e. the amount of money in the account satisfies the ODE

$$\frac{dy}{dt} = ky\,.$$

Verify that is $y = C\exp(kt)$ is a solution for $-\infty < t < \infty$.

- A **partial differential equation** (PDE) is an equation that relates values of a function of more than one variable (e.g. $y(x,t)$, say) to its partial derivatives, for example

$$\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial t \partial x} + 3\frac{\partial y}{\partial t} + xy = t^3 .$$

  Here $y(x,t)$ is the unknown function and x and t are the independent variables.

- The **order of a differential equation** is the order of the highest derivative of the unknown function.

Example 1.2 What is order of the following differential equations?

1. $\frac{dy}{dt} + 6y = 0$
2. $\frac{\partial^3 y}{\partial x^3} + \frac{\partial^2 y}{\partial t^2} + 3xy^6 = \frac{\partial^4 f}{\partial x^4}$, where f is a given function of x

- The nth order ordinary differential equation

$$\frac{d^n y}{dt^n} = f\left(t, y, \frac{dy}{dt}, \ldots, \frac{d^{(n-1)}y}{dt^{(n-1)}}\right)$$

  is said to be **linear** if f is a linear function of the variables $y, \frac{dy}{dt}, \ldots, \frac{d^{(n-1)}y}{dt^{(n-1)}}$, otherwise the differential equation is said to be **nonlinear**.

Example 1.3 Are the following differential equations linear or nonlinear?

1. $a_0(t)\frac{d^n y}{dt^n} + a_1(t)\frac{d^{n-1}y}{dt^{n-1}} + \cdots + a_n(t)y = g(t)$
2. $\frac{d^3 y}{dt^3} + 2e^t\frac{d^2 y}{dt^2} + y\frac{dy}{dt} = t^5$
3. $\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0$, where g and L are constants
4. $\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0$, where g and L are constants

2 First Order Ordinary Differential Equations

Reference: Boyce & DiPrima, Chapter 2

In this section we consider differential equations in the (standard) form

$$\boxed{\frac{dy}{dt} = f(t,y)\,,} \tag{2.1}$$

where f (the right-hand side) is a given function of the two variables t and y. Any differentiable function $y = \phi(t)$ that satisfies this equation for all t in some interval $I : \alpha < t < \beta$ is called a solution. Our objective is discuss whether such functions exist and if so to find the solution. There are different methods of finding the solution for different forms of the function f.

2.1 Linear Equations

When the function f depends linearly on the dependent variable y then equation (2.1) can be written in the form

$$\boxed{\frac{dy}{dt} + p(t)y = g(t)\,,} \tag{2.2}$$

where we assume that p and g are given continuous functions on the interval $I : \alpha < t < \beta$. The equation is known as a **first order linear differential equation**.

2.1.1 Integrating factor method[2]

[2] Integrating factors have been seen before in MTH1002, part 2

To solve equation (2.2) we first find the **integrating factor** $\mu(t)$ defined by

$$\boxed{\mu(t) = \exp\left(\int p(t)\,dt\right)\,.} \tag{2.3}$$

Note that $\mu(t)$ is positive for all t.

Then we note that differentiating $\mu(t)$ gives

$$\frac{d\mu}{dt} = \exp\left(\int p(t)\,dt\right)p(t) = \mu(t)p(t)\,. \tag{2.4}$$

We will use this below.

Now multiply (2.2) by $\mu(t)$ to obtain

$$\mu(t)\frac{dy}{dt} + \mu(t)p(t)y = \mu(t)g(t)\,. \tag{2.5}$$

and notice that this is the same equation as

$$\boxed{\frac{d}{dt}\left[\mu(t)y\right] = \mu(t)g(t)} \tag{2.6}$$

because we can expand the left-hand side of (2.6) using the product rule

$$\begin{aligned}\frac{d}{dt}\left[\mu(t)y\right] &= \mu(t)\frac{dy}{dt} + y\frac{d\mu}{dt}\\ &= \mu(t)\frac{dy}{dt} + y\mu(t)p(t) \quad \text{using (2.4).}\end{aligned}$$

Now we solve (2.6) by integrating both sides to obtain

$$\mu(t)y = \int \mu(t)g(t)\,dt + c\,,$$

where c is an arbitrary constant. Rearranging gives the **general solution**

$$y = \frac{\int \mu(t)g(t)\,dt + c}{\mu(t)}\,. \tag{2.7}$$

The graphs of $y(t)$ for a chosen constant c are called **integral curves**. They are the solutions of the first order linear differential equation (2.2). There is a infinite family of such curves, one for each c.

A first order differential equation together with an **initial condition**, written

$$\boxed{y(t_0) = y_0\,,}$$

is known as an **initial value problem** (IVP). The solution to an initial value problem is the one member of the infinite family of integral curves that passes through the point (t_0, y_0).

Example 2.1 Consider the initial value problem

$$\frac{dy}{dt} - 2y = e^{2t}, \quad y(0) = 2\,.$$

Plot the integral curves and find the solution of the IVP.

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2.2 Separable Equations[3]

[3] Separable equations have been seen before in MTH1002, part 2

Recall the first order ODE (2.1)

$$\frac{dy}{dt} = f(t,y)\,.$$

If $f(t,y)$ can be separated into a function of t only ($F(t)$, say) multiplied by a function of y only ($G(y)$, say) then we have a **separable first order ODE**

$$\boxed{\frac{dy}{dt} = F(t)G(y(t))} \tag{2.8}$$

2.2.1 Special solutions

If $G(y_*) = 0$ then $y(t) = y_*$ is a **special solution**.

2.2.2 Typical solutions

If $G \neq 0$ then we may divide by it and find the **typical solutions** of

$$\frac{1}{G(y)}\frac{dy}{dt} = F(t).$$

Integrating with respect to t gives

$$\int \frac{1}{G(y)}\frac{dy}{dt}dt = \int F(t)dt + c,$$

where c is a constant. Now we take y as the integration variable on the left-hand side to obtain

$$\int \frac{1}{G(y)}dy = \int F(t)dt + c.$$

2.2.3 General solution

The general solution is the union of the set of all solutions.

Example 2.2 Find the solution of the initial value problem

$$\frac{dy}{dt} = \frac{y\cos t}{1+2y^2}, \quad y(0) = 1\,.$$

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2.2.4 Test for separability

Note that if $f(t,y) = F(t)G(y)$ for all t and y then

$$f(a,b)f(c,d) = F(a)G(b)F(c)G(d) = f(a,d)f(c,b)$$

for all a, b, c and d. Hence, if you can find one set of a, b, c and d for which this equality does not hold, then you know that the ODE is not separable.

Activity 2.3 Is the ODE

$$y^2\frac{dy}{dt} + t^2 = ty^2$$

separable?

2.3 Homogeneous Equations

The first order ODE (2.1)

$$\frac{dy}{dt} = f(t,y)\,,$$

is said to be homogeneous whenever the function f does not depend on t and y separately, but only depends on their ratio y/t or t/y. Thus, homogeneous first order ODEs take the form

$$\boxed{\frac{dy}{dt} = H\left(\frac{y}{t}\right)} \tag{2.9}$$

The form of equation (2.9) suggests that the equation can be simplified by introducing a new variable to represent the ratio. Thus we set

$$v = \frac{y}{t}\,, \quad t \neq 0\,.$$

Then, using the product rule,

$$y = vt \quad \Rightarrow \frac{dy}{dt} = v + t\frac{dv}{dt}$$

and so substituting into (2.9) gives the ODE

$$v + t\frac{dv}{dt} = H(v)$$

which is of separable form

$$\int \frac{dt}{t} = \int \frac{dv}{H(v) - v}$$

for $H(v) \neq v$.

Example 2.4 Solve the differential equation

$$t\frac{dy}{dt} = t + y\,.$$

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2.3.1 Test for homogeneity

A homogeneous equation has

$$f(wt, wy) = f(t, y), \quad \forall\, w, t, y\,.$$

Hence, if you can find a set of w, t, y for which this equality does not hold, then you know that the ODE is not homogeneous.

Important note: The classification 'homogeneous' is used in more than one way in the study of differential equations. With experience, you will be able to distinguish between these.

2.4 Existence & Uniqueness

In the previous sections we have used a variety of methods to find solutions to the initial value problem

$$\frac{dy}{dt} = f(t,y), \quad y(t_0) = y_0\,. \tag{2.10}$$

In this section we address questions of a more general nature, such as:

- Does an initial value problem always have a solution?
- Can it have more than one solution?
- Is the solution valid for all t, or only for some restricted interval near the initial point t_0?

First, we require the definition of a continuous function:[4]

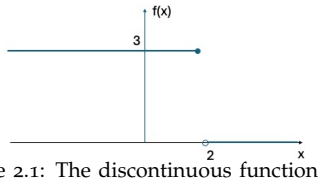


Figure 2.1: The discontinuous function $f(x) = 3$ for $x \leq 2$ and $f(x) = 0$ for $x > 2$.

[4] see the lecture notes for MTH1002, part 1. Note that if the left-hand and right-hand limits are different

$$\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$$

then

$$\lim_{x\to a} f(x) \quad \text{does not exist.}$$

For example, the function defined by

$$f(x) = \begin{cases} 3 & x \leq 2 \\ 0 & x > 2 \end{cases}$$

is not continuous at $x = 2$ (see figure 2.1). Also, if a function is not defined at a certain point then it cannot be continuous at that point. For example,

$$f(x) = \frac{1}{x-3}$$

is not defined at $x = 3$, see figure 2.2.

Definition 2.5 A function $f(x)$ is **continuous** at $x = a$ if

$$\lim_{x\to a} f(x) = f(a)\,.$$

A function is continuous on an interval if it is continuous at all points in that interval.

2.4.1 Linear first order ODEs

Theorem 2.6 If the functions $p(t)$ and $g(t)$ are continuous on the open interval $I : \alpha < t < \beta$, with $t_0 \in I$, then there exists a unique function $y = \phi(t)$ that satisfies the linear differential equation

$$\frac{dy}{dt} + p(t)y = g(t) \tag{2.11}$$

for all t in I, with $y(t_0) = y_0$, where y_0 is an arbitrary prescribed value.

Note that this theorem asserts both existence and uniqueness of the solution of the initial value problem (i.e. we are told that the IVP has a solution and it has only one solution). The theorem also states that the solution exists throughout the interval I on which p and g are continuous and that contains t_0.

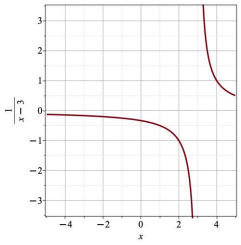


Figure 2.2: The discontinuous function $f(x) = 1/(x-3)$ that is undefined at $x = 3$.

2.4.2 First order ODEs

Definition 2.7 In one-dimensional space, a **neighbourhood** of a point $t \in \mathbb{R}$ is any (possibly very small) open interval containing the point t. Similarly, in two-dimensional space, a neighbourhood of a point $(t, y) \in \mathbb{R}^2$ is any open set (e.g. a rectangle or a disc) containing the point (t, y). A certain property of a function f is said to be a **local property**, or is **locally satisfied**, if it is satisfied in a neighbourhood of the point.

Theorem 2.8 — Peano's existence theorem. Let the function $f(t, y)$ be continuous in a neighbourhood of (t_0, y_0). Then the IVP

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

is **solvable locally**, i.e. it has **at least one solution** in a neighbourhood of t_0.

Note that the interval of t where the solution exists may be smaller than the interval over which f is continuous.

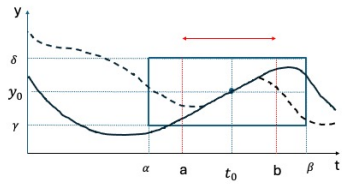


Figure 2.3: Peano's existence interval.

Theorem 2.9 — Picard's existence & uniqueness theorem. Let the functions $f(t, y)$ and its partial derivative $\partial f/\partial y$ be continuous in a neighbourhood of (t_0, y_0). Then the IVP

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

is **uniquely solvable locally**, i.e. there is **exactly one solution** in a neighbourhood of t_0.

Note that the interval of t over which the solution is unique may be smaller than the interval over which f and $\partial f/\partial y$ are continuous.

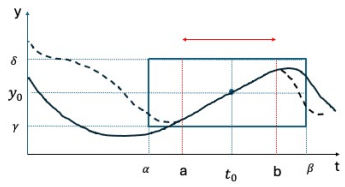


Figure 2.4: Picard's uniqueness interval.

Example 2.10 Without solving the IVP

$$t\frac{dy}{dt} + 2y = 4t^2, \quad y(1) = 2,$$

find the interval in which it has a unique solution. Then solve the IVP to find the solution in that interval.

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Example 2.11 Find the interval on which the IVP

$$\frac{dy}{dt} + (\tan t)y = \sin t, \quad y(\pi) = 0$$

is guaranteed to have a unique solution.

We note that questions concerning existence and uniqueness of solutions are different to the question of actually finding a solution. One may ask why we are interested in existence and uniqueness. One reason is that in a mathematical modelling problem you may find a solution but it might not correspond to the actual behaviour of the real physical system if the solution isn't unique; i.e. you have found a solution but not the one of physical interest. Alternatively, if you can show that the solution exists and is unique, then, even if you cannot find the solution analytically, you can proceed to find an approximation to it computationally.